

Fur and hair

Invertebrates sometimes look furry, but only mammals have real fur and hair. A fur coat protects against the heat and the cold, and it often works as camouflage.

On some mammals, fur can be amazingly dense—sea otters have about 1 million hairs on every square inch (150,000 hairs per sq cm) of their bodies, which is as many as most people have on their entire heads. Hairs grow in small pits, or follicles, in the skin. Each follicle contains a tiny muscle and when this contracts, an animal's fur stands up on end.

Changing coats

Many woodland mammals change the color of their coats as they grow up. Wild boar piglets are born with brown-and-yellow-striped coats, which fade to a darker brown by the time they are five months old. Young deer, or fawns, have spotted coats at birth. In many species, these disappear by the time a fawn is one year old. The stripes and spots act as camouflage, helping to conceal the young in the sun-dappled shade under the trees.



Long guard hairs form coat surface



Close-up of fur layers

Underfur traps air, conserving warmth

Winter layers

Huddling together on the Alaskan tundra, these hardy musk oxen are wearing their warm winter coats. The dark-brown, outer fur layer is made up of extra-long guard hairs (see p. 122), which protect the oxen from the snow and gale-force winter winds. Beneath this coarse, shaggy layer grows a shorter, woolly undercoat called underfur—one of the lightest and warmest in the animal world. Like many hoofed mammals, musk oxen molt their coats twice a year and grow a shorter, cooler coat for the summer months.

Piglet's coloring contrasts with mother's dark, bristly coat

